

Visual and tactile motion cues enhance the categorisation of novel object shapes

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Figure S1 – Predicted categorisation accuracy (on the logit scale) and odds ratios for accuracy by Cue Condition and Exemplar Type for the Experiment 1A and 1B.

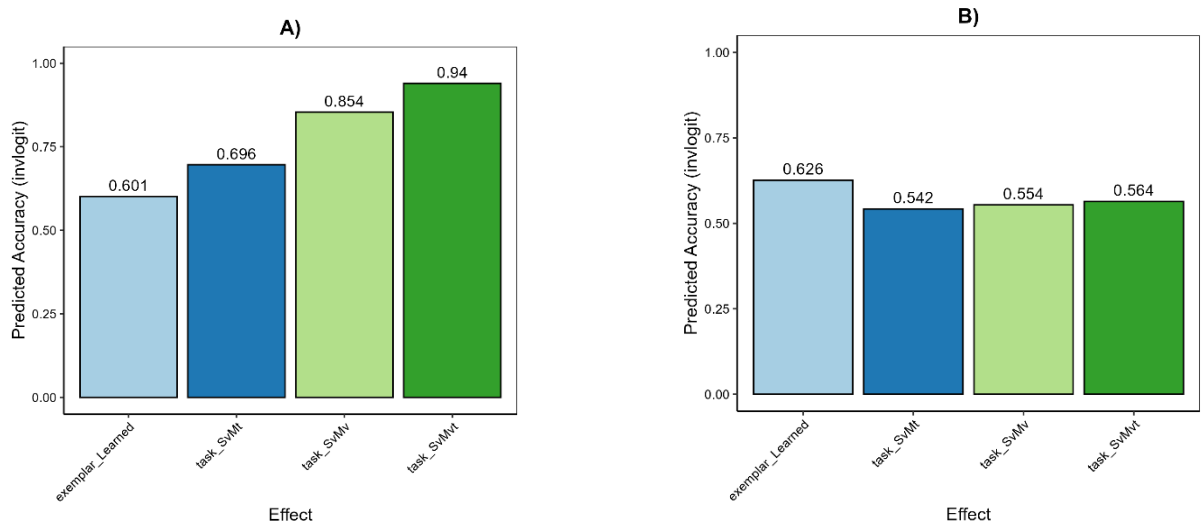


Figure S1.1 - Predicted categorisation accuracy by cue condition and exemplar type for Experiment 1A (Panel A) and Experiment 1B (Panel B), based on a binomial mixed-effects model. Accuracy is shown separately for learned and novel exemplars across four cue conditions (Sv, SvMv, SvMt and SvMvt). In the blocked presentation (1A), accuracy increases with the number of cues, indicating a multisensory benefit. In the interleaved presentation (1B), overall accuracy is lower and shows minimal variation across conditions. Points represent model-estimated accuracy with 95% confidence intervals.

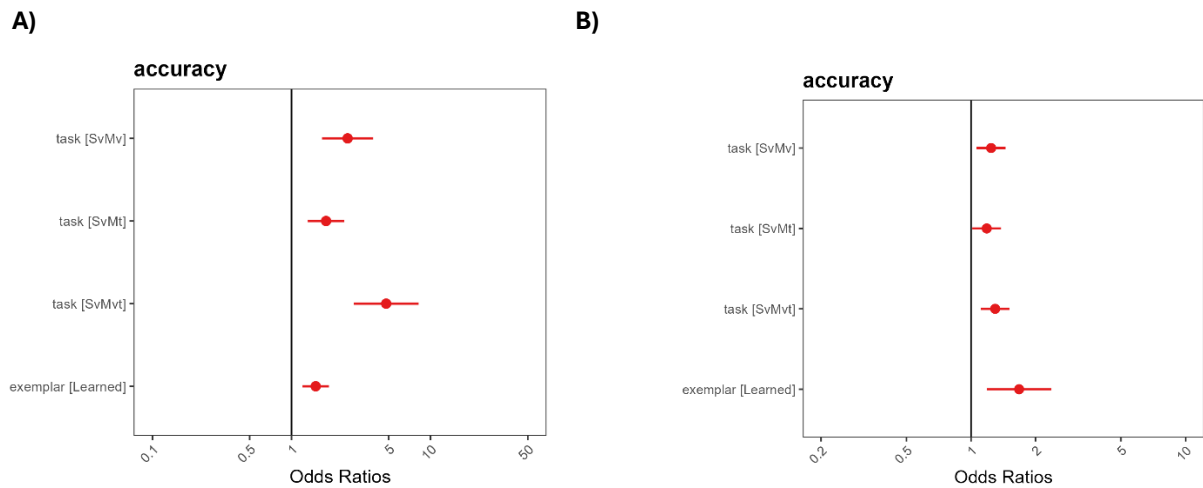


Figure S1.2 - Odds ratios for categorisation accuracy across cue conditions and exemplar types for Experiment 1A (Panel A) and Experiment 1B (Panel B), derived from a binomial mixed-effects model. Points indicate estimated odds ratios; horizontal lines show 95% confidence intervals. The vertical reference line at OR = 1 reflects chance-level performance. In the blocked condition (1a), odds of correct categorisation increase with additional cues. In the interleaved condition (1b), cue effects are attenuated.

Figure S2 – Predicted categorisation accuracy (on the logit scale) and odds ratios for accuracy by Cue Condition and Exemplar Type for the Experiment 2A and 2B.

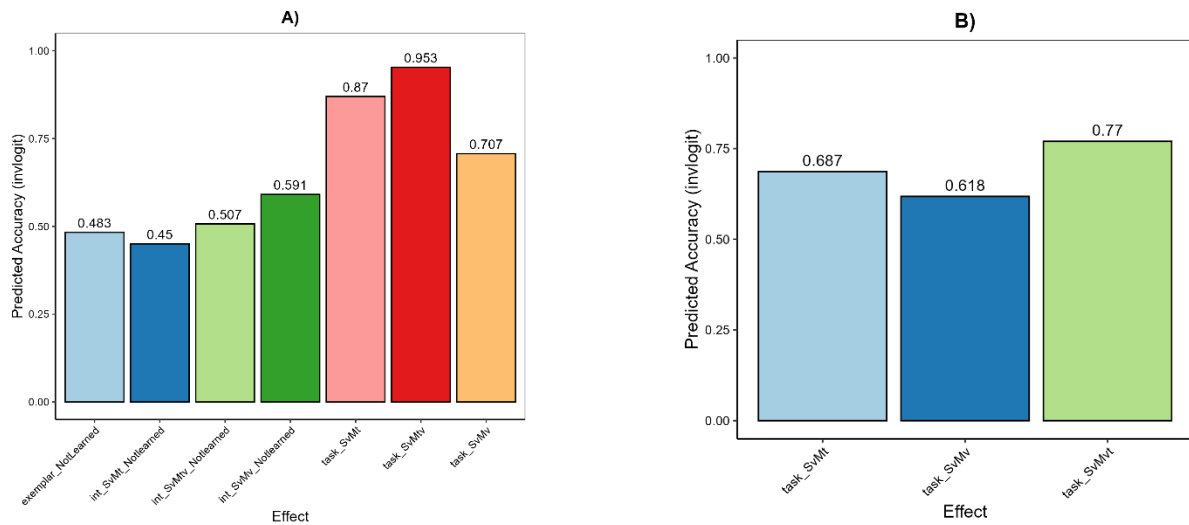


Figure S2.1- Predicted categorisation accuracy by cue condition and exemplar type for Experiment 2A (Panel A, blocked presentation) and Experiment 2B (Panel B, interleaved presentation), based on a binomial mixed-effects model. In Experiment 2A, accuracy increases with additional predictive cues, with the highest accuracy observed in the SvMvt condition, while performance in the Sv condition is substantially lower, particularly for novel exemplars. In Experiment 2B, accuracy also improves across conditions but shows reduced sensitivity to cue informativeness, suggesting that interleaved presentation may limit the benefit of multisensory integration.

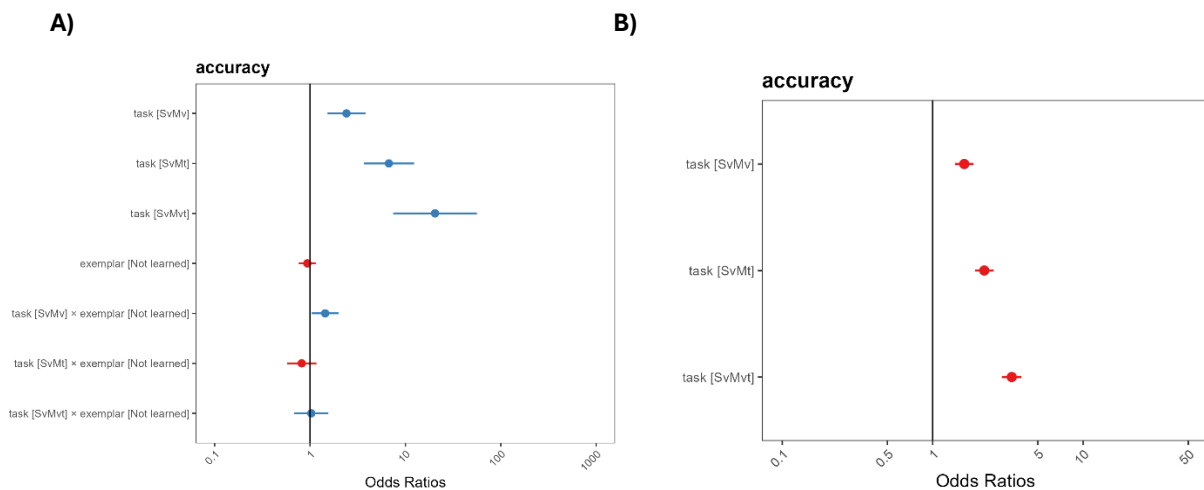


Figure S2.2 - Odds ratios for categorisation accuracy across cue conditions and exemplar types in Experiment 2A (Panel A) and Experiment 2B (Panel B), derived from binomial mixed-effects models. Each point represents the estimated odds ratio for a given condition, and horizontal lines indicate 95% confidence intervals. The vertical dashed line at OR = 1 marks the reference value (no difference). In Experiment 2A, odds ratios increase with additional cues, especially for novel exemplars. In Experiment 2B, cue effects are present but attenuated.

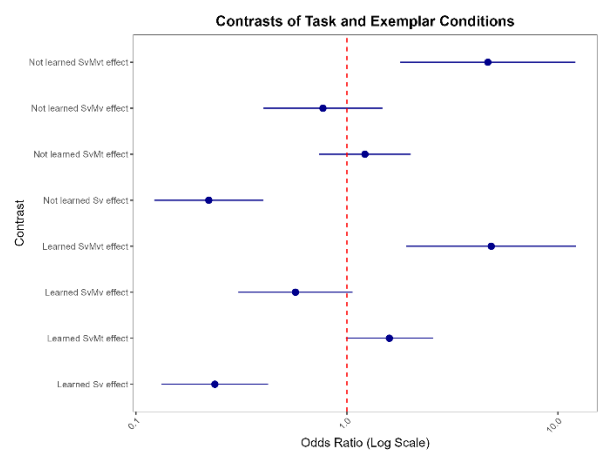
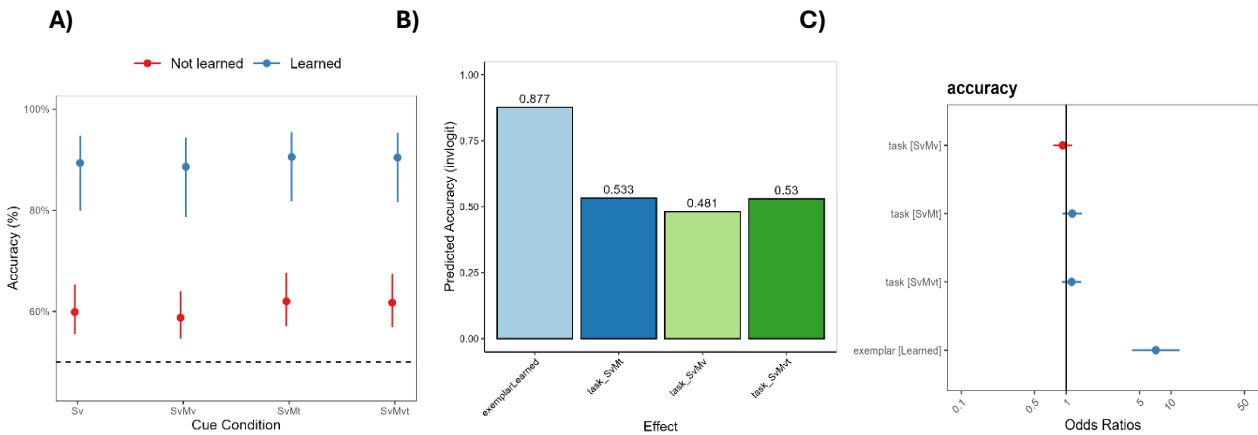


Figure S2.3 - Estimated odds ratios for contrasts between cue conditions and exemplar types in Experiment 2A, plotted on a logarithmic scale. Each point reflects the odds ratio for a specific contrast comparing categorisation accuracy across cue and exemplar combinations (S_v , S_vM_v , S_vM_t and S_vM_{vt}), with horizontal lines denoting 95% confidence intervals. The red dashed line at OR = 1 represents the baseline (no difference). Contrasts are displayed separately for learned and novel exemplars. Odds ratios above 1 indicate improved performance relative to baseline; confidence intervals not crossing the reference line suggest statistically significant effects.

Figure S3 –Predicted categorisation accuracy (on the logit scale) and odds ratios for accuracy by Cue Condition and Exemplar Type for Experiment 3.



Predicted categorisation accuracy by cue condition (S_v , S_vM_v , S_vM_t and S_vM_{vt}) and exemplar type (learned vs. novel) in Experiment 3. Panel A shows accuracy estimates (%) indicating consistently high performance for learned exemplars across all conditions, with lower and less responsive accuracy for novel exemplars, suggesting limited generalisation and minimal benefit from untrained cues. Panel B presents accuracy on the logit scale, showing that cue condition had little impact on performance, particularly for novel exemplars. Panel C displays odds ratios with 95% confidence intervals from the mixed-effects model; values above 1 reflect improved accuracy relative to baseline, with the vertical line at OR = 1 indicating no effect.

Table S1*Model syntax and key for the final models fit across Experiments 1A, 1B, 2A, 2B and 3.*

Model terms	Key
Experiment 1A	
Model1<- glmer(accuracy ~ 1 + task + exemplar + (1 + task + exemplar ID), data = acc_data, family = binomial(link = "logit"), control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5)))	Accuracy performance (binomial; 1,0) Cue condition main effect Exemplar type main effect Random effects; random intercept + random Linear trial slope Data frame + model optimiser.
Experiment 1B	
Model1<- glmer(accuracy ~ 1 + task + exemplar + (1 + exemplar ID), data = acc_data, family = binomial(link = "logit"), control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5)))	Accuracy performance (binomial; 1,0) Cue condition main effect Exemplar type main effect Random effects; random intercept + random Linear trial slope Data frame + model optimiser.
Experiment 2A	
Model2<- glmer(accuracy ~ 1 + task + exemplar + task:exemplar + (1 + task ID), data = acc_data, family = binomial(link = "logit"), control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5)))	Accuracy performance (binomial; 1,0) Cue condition main effect Exemplar type main effect Cue condition * Exemplar type interaction effect Random effects; random intercept + random Linear trial slope Data frame + model optimiser.
Experiment 2B	
Model4<- glmer(accuracy ~ 1 + task + (1 ID), data = acc_data, family = binomial(link = "logit"), control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5)))	Accuracy performance (binomial; 1,0) Cue condition main effect Random effects Data frame + model optimiser.
Experiment 3	
Model14 <- glmer(accuracy ~ 0 + task + exemplar + (1+ exemplar ID), data = acc_data, family = binomial(link = "logit"), control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5)))	Accuracy performance (binomial; 1,0) Cue condition main effect Exemplar type main effect Random effects; random intercept + random Linear trial slope Data frame + model optimiser.

VISUAL AND TACTILE MOTION CUES ENHANCE CATEGORISATION – Model Information (SMI)

Table S2*Model fit indices across models in Experiments 1A, 1B, 2A, 2B and 3. The final model selected is in bold.*

Experiment 1A			
	Likelihood ratio test	AIC	BIC
Model 1 (without interaction)		9712.9	9857.0
Model 2 (with interaction)	$\chi^2(3) = 6.40, p = 0.094$	9712.5	9878.2
Model 1			
R2 (conditional)	R2 marginal	ICC	RMSE
0.449 (44.9%)	0.060 (6.0%)	0.389	0.393
Experiment 1B			
	Likelihood ratio test	AIC	BIC
Model 1 (without interaction)		7686.8	7741.3
Model 2 (with interaction)	$\chi^2(3) = 1.47, p = 0.689$	7766.3	7766.3
Model 1			
R2 (conditional)	R2 marginal	ICC	RMSE
0.327 (32.7%)	0.015 (1.5%)	0.31	0.440
Experiment 2A			
	Likelihood ratio test	AIC	BIC
Model 1 (without interaction)		6273.4	6376.7
Model 2 (with interaction)	$\chi^2(3) = 9.09, p = .028$	6270.3	6394.2
Model 2			
R2 (conditional)	R2 marginal	ICC	RMSE
0.561 (56.1%)	0.165 (16.5%)	0.396	0.367
Experiment 2B			
	Likelihood ratio test	AIC	BIC
Model 2 (with interaction)	$\chi^2(3) = 0.733, p = 0.866$	8345.8	8407.3
Model 1 (without interaction)		8340.5	8381.5
Model 3 (no interaction – exemplar main effect)	$\chi^2(1) = 0.953, p = 0.329$	8339.5	8373.6
Model 4 (no interaction – task main effect)	$\chi^2(3) = 276.7, p < 0.001$	8611.2	8631.7
Model 4			
R2 (conditional)	R2 marginal	ICC	RMSE
0.165 (16.5%)	0.49 (49%)	0.114	0.455
Experiment 3			
	Likelihood ratio test	AIC	BIC
Model 2 (with interaction)	$\chi^2(3) = 2.75, p = 0.432$	4537.7	4613.1
Model 1 (without interaction)		4534.5	4589.3
Model 3 (no interaction – task main effect)	$\chi^2(3) = 5.251, p = 0.154$	4533.7	4568.0
Model 4 (no interaction – exemplar main effect)	$\chi^2(1) = 34.95, p < 0.001$	4567.4	4615.4
Model 4			
R2 (conditional)	R2 marginal	ICC	RMSE
0.121 (12.1%)	0.551 (55.1%)	0.431	0.312

Table S3

Predicted mean accuracy values per cue condition and exemplar type during test (95% confidence intervals).

Experiment 1A	
Cue condition	
S_v	0.65 (0.56, 0.72)
S_vM_v	0.82 (0.74, 0.88)
S_vM_t	0.76 (0.70, 0.82)
S_vM_{vt}	0.90 (0.83, 0.94)
Exemplar type	
Learned	0.82 (0.75, 0.88)
Not learned	0.75 (0.69, 0.81)
Experiment 1B	
Cue condition	
S_v	0.69 (0.59, 0.77)
S_vM_v	0.73 (0.64, 0.80)
S_vM_t	0.72 (0.63, 0.80)
S_vM_{vt}	0.74 (0.65, 0.81)
Exemplar type	
Learned	0.76 (0.65, 0.84)
Not learned	0.65 (0.59, 0.70)
Experiment 2A	
Learned Exemplar x Cue condition	
S_v	0.61 (0.56, 0.66)
S_vM_v	0.79 (0.69, 0.86)
S_vM_t	0.91 (0.85, 0.95)
S_vM_{vt}	0.97 (0.92, 0.99)
Not Learned Exemplar x Cue condition	
S_v	0.60 (0.54, 0.65)
S_vM_v	0.84 (0.75, 0.90)
S_vM_t	0.89 (0.89, 0.94)
S_vM_{vt}	0.97 (0.92, 0.99)
Experiment 2B	
Cue condition	
S_v	0.52 (0.45, 0.58)
S_vM_v	0.63 (0.57, 0.69)
S_vM_t	0.70 (0.65, 0.75)
S_vM_{vt}	0.78 (0.73, 0.82)
Experiment 3	
Exemplar type	
Learned	0.96 (0.91, 0.98)
Not learned	0.76 (0.70, 0.81)

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Figure S4.1- Beta coefficients and 95% confidence intervals for the results of Experiment 1A. From left to right: Model with two-way interactions, model with main effects only.

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	0.76	0.31, 1.2	<0.001
SvMt	0.56	0.22, 0.90	0.001
SvMvt	1.6	1.0, 2.2	<0.001
exemplar			
Not learned	—	—	
Learned	0.35	0.08, 0.61	0.010
task * exemplar			
SvMv * Learned	0.29	0.00, 0.59	0.049
SvMt * Learned	0.02	-0.26, 0.29	>0.9
SvMvt * Learned	-0.09	-0.40, 0.22	0.6

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	0.93	0.51, 1.4	<0.001
SvMt	0.57	0.27, 0.87	<0.001
SvMvt	1.6	1.0, 2.1	<0.001
exemplar			
Not learned	—	—	
Learned	0.40	0.18, 0.62	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Figure S4.2 - Beta coefficients and 95% confidence intervals for the results of Experiment 1B. From left to right: Model with two-way interactions, model with main effects only.

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	0.31	0.07, 0.56	0.012
SvMt	0.26	0.01, 0.50	0.038
SvMvt	0.29	0.04, 0.53	0.021
exemplar			
Not learned	—	—	
Learned	0.61	0.21, 1.0	0.003
task * exemplar			
SvMv * Learned	-0.16	-0.48, 0.15	0.3
SvMt * Learned	-0.15	-0.46, 0.17	0.4
SvMvt * Learned	-0.05	-0.36, 0.27	0.8

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	0.22	0.06, 0.37	0.006
SvMt	0.17	0.01, 0.32	0.032
SvMvt	0.26	0.10, 0.41	0.001
exemplar			
Not learned	—	—	
Learned	0.52	0.17, 0.86	0.004

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Figure S5.1 - Beta coefficients and 95% confidence intervals for the results of Experiment 2A. From left to right: Model with two-way interactions, model with main effects only.

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	0.88	0.42, 1.3	<0.001
SvMt	1.9	1.3, 2.5	<0.001
SvMvt	3.0	2.0, 4.0	<0.001
exemplar			
Learned	—	—	
Not learned	-0.07	-0.27, 0.14	0.5
task * exemplar			
SvMv * Not learned	0.37	0.04, 0.69	0.026
SvMt * Not learned	-0.20	-0.55, 0.15	0.3
SvMvt * Not learned	0.03	-0.38, 0.44	0.9

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	1.0	0.56, 1.4	<0.001
SvMt	1.8	1.2, 2.4	<0.001
SvMvt	3.0	2.0, 4.0	<0.001
exemplar			
Learned	—	—	
Not learned	0.00	-0.13, 0.13	>0.9

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Figure S5.2 - Beta coefficients and 95% confidence intervals for the results of Experiment 2B. From left to right: Model with two-way interactions, model with main effects only.

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	0.48	0.31, 0.66	<0.001
SvMt	0.79	0.61, 0.97	<0.001
SvMvt	1.3	1.1, 1.4	<0.001
exemplar			
Learned	—	—	
Not learned	-0.03	-0.23, 0.18	0.8
task * exemplar			
SvMv * Not learned	-0.01	-0.30, 0.29	>0.9
SvMt * Not learned	0.00	-0.30, 0.30	>0.9
SvMvt * Not learned	-0.12	-0.43, 0.19	0.5
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	0.48	0.34, 0.62	<0.001
SvMt	0.79	0.64, 0.93	<0.001
SvMvt	1.2	1.1, 1.4	<0.001
exemplar			
Learned	—	—	
Not learned	-0.05	-0.16, 0.05	0.3
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	0.48	0.34, 0.62	<0.001
SvMt	0.79	0.64, 0.93	<0.001
SvMvt	1.2	1.1, 1.4	<0.001
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

Figure S6 - Beta coefficients and 95% confidence intervals for the results of Experiment 3. From left to right: Model with two-way interactions, model with main effects only.

Characteristic	log(OR)	95% CI	p-value
task			
Sv	—	—	
SvMv	-0.08	-0.28, 0.13	0.5
SvMt	0.13	-0.08, 0.35	0.2
SvMvt	0.12	-0.09, 0.33	0.3
exemplar			
Not learned	—	—	
Learned	2.0	1.4, 2.5	<0.001
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			

Characteristic	log(OR)	95% CI	p-value
exemplar			
Not learned	—	—	
Learned	2.0	1.4, 2.5	<0.001
Abbreviations: CI = Confidence Interval, OR = Odds Ratio			